	Case Name: Active Recreation Construction Project Owner: Eder Omar Vilca Carrillo	Sector	Construction
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources Schedule with costs
		Risk Analysis	Random simulation One of nine std. scenarios User defined distributions
Processed by	**Zahra Hussaini**		
Date	**2025**	Project Control	Automatic tracking
File Name	**C2025-23**		Tracking based on user input

1. Project description

Project authenticity

This project describes the various construction, safety, and infrastructure activities involved in the development of an **active recreation area** at Tintaya Park, located in the Tintaya Workers Urbanization, District and Province of Arequipa, including preliminary works, temporary facilities, sanitary services, landscaping, paving, and electrical installations. The project consists of activity and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	55
Planned Duration (PD)	60 days
Budget At Completion (BAC)	124330.97 EUR
Renewable Resources	9 Types
Consumable Resources	0 Types

Network topology	
Serial/Parallel (SP)	38%
Activity Distribution (AD)	61%
Length of Arcs (LA)	0%
Topological Float (TF)	68%

* standard eight-hour working days

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	13.3	13.1	1.9
CRI-rho	13.3	13.1	2.0
CRI-tau	12.2	9.5	1.2

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	4.7	14.1	6.0
SI	20.9	27.2	1.4
SSI	4.7	10.9	2.2
CRI-r	10.9	8.5	1.3
CRI-rho	10.3	8.3	1.3
CRI-tau	11.9	6.9	0.0

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	14.0	10.7	3.8
CRI-rho	14.0	10.7	3.8
CRI-tau	14.0	10.7	3.8

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section “2.2. Risk Analysis”. Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	26.4	26.4
PV - SPI	26.9	26.9
PV - SCI	28.5	28.5
ED - 1	22.1	22.1
ED - SPI	22.7	22.7
ED - SCI	23.3	23.3
ES - 1	22.5	22.5
ES - SPI(t)	23.8	23.8
ES - SCI(t)	24.4	24.4

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	2.1	1.1
CPI	2.2	2.0
SPI	2.2	1.8
SPI(t)	2.8	2.6
SCI	2.8	2.6
SCI(t)	3.6	3.4
0.8 CPI + 0.2 SPI	2.2	1.9
0.8 CPI + 0.2 SPI(t)	2.2	2.1

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 1 tracking period. The Real Duration and Real Cost mentioned in section “2.3.3. Earned Value Management” are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	N/A	N/A	N/A	N/A	N/A	N/A	N/A
std dev	N/A	N/A	N/A	N/A	N/A	N/A	N/A
final	0	0	-2	1.0	1	0.98	1

2.3.3.2. Time forecasting

PD	60	Real Duration	59	Early	1%

EAC(t)			**Real Accuracy**	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	N/A	N/A	N/A	N/A
PV - SPI	N/A	N/A	N/A	N/A
PV - SCI	N/A	N/A	N/A	N/A
ED - 1	N/A	N/A	N/A	N/A
ED - SPI	N/A	N/A	N/A	N/A
ED - SCI	N/A	N/A	N/A	N/A
ES - 1	N/A	N/A	N/A	N/A
ES - SPI(t)	N/A	N/A	N/A	N/A
ES - SCI(t)	N/A	N/A	N/A	N/A

2.3.3.3. Cost forecasting

BAC	124330.97 €	Real Cost	124330.97 €	On budget	0%

EAC			**Real Accuracy**	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	N/A	N/A	N/A	N/A
CPI	N/A	N/A	N/A	N/A
SPI	N/A	N/A	N/A	N/A
SPI(t)	N/A	N/A	N/A	N/A
SCI	N/A	N/A	N/A	N/A
SCI(t)	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI	N/A	N/A	N/A	N/A
0.8 CPI + 0.2 SPI(t)	N/A	N/A	N/A	N/A